IT45

User Guide

Multi-venue funding trader: Hyperliquid, Arcus, Bulk, Reya, Meridian

IT45 is your multi-venue trading dashboard: it watches funding on Hyperliquid, Arcus, Bulk, Reya and Meridian, trades funding spreads automatically, and lets you place orders yourself. This build trades with play money only (paper or testnet); mainnet arrives with M3. Rest the pointer on anything in the dashboard for the short version of this guide.

Build	M2.x, paper and testnet only
Edition	4 October 2026
Prepared for	Iman

Hover help in the dashboard is the short form of this guide and always matches the running build.

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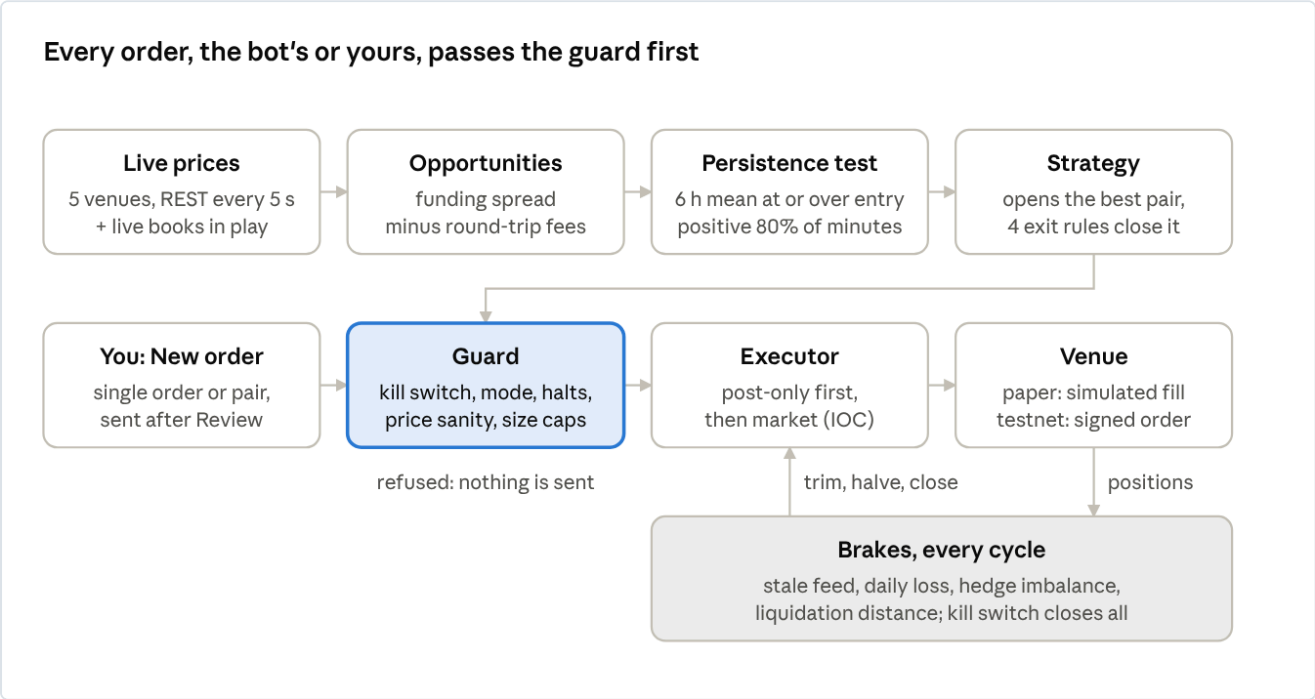
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1 What IT45 does

IT45 earns funding, not price bets: it holds the same asset long on one venue and short on another, so price moves cancel and the funding difference is the income.

- **Cross-venue funding arb:** long where funding is cheap, short where it is rich, on any two of the five venues.
- **Hyperliquid spot-perp carry:** long Hyperliquid spot, short the Hyperliquid perp, collecting funding while it stays positive.

A worked example. BTC funding is 5%/yr on one venue and 20%/yr on another, so the pair earns 15%/yr on each leg's notional, about 0.17 bp an hour. Opening and closing both legs as taker on Hyperliquid (4.5 bp) and Bulk (3.5 bp) costs 2 × 8 = 16 bp, paid back after about 93 hours. As maker on both (1.5 bp and 0 bp) it costs 3 bp, paid back in about 18 hours. That is why IT45 tries post-only orders first, and why it only trades spreads that have lasted.



How an order travels, from price feed to venue. Every order passes the guard; the brakes watch positions on their own schedule.

The top row is how the bot finds a trade; the bottom row is how any order, the bot's or yours, reaches a venue. Nothing skips the guard, and the brakes watch every position on their own schedule.

2 Start, stop and update

One PowerShell command starts everything. The dashboard then opens at http://127.0.0.1:8045 in your browser.

```
cd "D:\PROJECTS\INTELLIGENT TRADER\IT45"
.\run.ps1
```

- **First run** creates `.venv` and installs the Python packages (a few minutes). Later runs start in seconds.
- **Stop:** press Ctrl+C in the PowerShell window. Closing the browser tab does not stop IT45; the bot keeps running while the window is open.
- **Options:** `.\run.ps1 --no-browser` starts without opening a tab. `.\run.ps1 --port 9000` uses another port.
- **Update to the newest build:** stop IT45, run `git pull`, then `.\run.ps1` again. The built dashboard is in the repo, so you never need Node.
- **If PowerShell refuses to run scripts:** run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once.
- **If the install fails,** IT45 does not start and shows the pip error in red. Send that text to Claude.

Your data lives in the `data` folder: `it45.db` (prices, funding history, orders, fills, pairs) and `settings.json` (everything you set in the dashboard). Both are git-ignored and survive updates. Keys live in `.env`, never in the dashboard.

3 Finding your way

The left menu has nine sections; click one to open its pages. Every page has a title and one line saying what it is for.

Section	Pages	Use it to
Overview	Summary, Venue health	See at a glance whether data is flowing and where the best spreads are
Venues	Hyperliquid, Arcus, Bulk, Reya, Meridian	Inspect one exchange: feed, fees, every market
Markets	Funding matrix, Price matrix, Hyperliquid spot	Compare one asset across venues
Opportunities	Cross-venue funding, Spot-perp carry	Rank spreads net of fees
Strategies	Funding arb, Spot carry	Start or stop the bot, see its candidates, edit its rules
Orders & positions	New order, Positions, Open orders, Trade history	Trade yourself and see what is held and what happened
Risk	Limits, Brakes, Kill switch	Decide where and how much the bot may trade; stop everything
Logs	Events	Read feed failures, recoveries and settings changes
Settings	General, Fees, Venues, API keys	Polling, fee overrides, venue feeds, key status

Top bar, left to right:

- **Breadcrumb** shows the section and page you are on.
- **Venue dots:** green = data is live, amber = stale, red = down or switched off. Click one to open that venue's page. Hover a dot for its age and last error.
- **Mode pill** summarises how IT45 is trading right now, for example "Paper, bot stopped". It lists the modes of the enabled venues ("Paper + Testnet") and whether a strategy is running, and turns red when the kill switch is engaged. Click it to go to the kill switch page.
- **? button** turns hover help on and off.
- **Sun/moon button** switches dark and light themes. Your choice is remembered on this computer.

Hover help is the built-in manual. Rest the pointer on any label, column header, number, button or setting for half a second and a plain-English explanation appears. A table cell explains its column. Every visible element has one (checked automatically: 0 of 1,964 missing).

Collapse at the bottom of the menu narrows it to icons for more table space. The address bar keeps the page (for example `#/orders/new`), so you can bookmark pages.

4 Overview and Venues

Overview tells you in five seconds whether IT45 is healthy. Venues shows one exchange in full.

Overview → Summary

- **Venue feeds live:** venues that answered their last poll within the stale limit. 5/5 is normal.
- **Perp markets tracked:** perpetual markets read from all venues together.
- **Best net funding spread:** the top cross-venue spread after fees, if today's rates held. A reason to look, not a forecast.
- **Stored snapshots:** rows of price and funding history in `data/it45.db`. The strategies need several hours of this before they trade.
- **Venue cards and Top cross-venue funding spreads:** the eight best spreads right now; click through to Opportunities for all of them.

Overview → Venue health has one row per venue: State (live, stale, down, disabled), Latency, Last update, Failed polls in a row, Last error, and **Live book**. Live book shows the websocket subscriptions: IT45 opens a live order-book stream only for markets it is working on (open pairs, resting orders, top candidates, the market open in New order). "Idle" means nothing is needed from that venue right now; that is normal.

Venues → one page per exchange shows the feed state, base fees, your account (only in testnet mode with a key), and a table of every market: price, mark, oracle, funding per hour and per year, open interest, max leverage, 24-hour volume and status. Filter by name, and switch between perps and spot on Hyperliquid.

Venue	Base maker	Base taker	What is special
Hyperliquid	1.5 bp	4.5 bp	Only venue with spot, so the only home of spot-perp carry; testnet connector
Arcus	0 bp	2.25 bp	Crypto plus stock, index and commodity perps; paper only for now
Bulk	0 bp	3.5 bp	Testnet connector verified end to end
Reya	0 bp	3 bp	Publishes funding in percent per hour; IT45 converts; paper only
Meridian	0 bp	0.5 bp	Perps on Robinhood Chain, including gold, silver, SPY, QQQ; testnet connector

1 bp (basis point) = 0.01%. These are the lowest volume tiers; if yours are better, enter them under Settings → Fees. All five venues settle funding hourly, and IT45 shows funding per year so the venues compare directly.

5 Markets and Opportunities

Markets compares one asset across venues; Opportunities ranks the spreads you could trade, after fees. Both are screens: every number assumes today's funding rate holds.

Markets → Funding matrix lists every asset listed on two or more venues, one column per venue, funding per year. Cells are shaded against that asset's own median across venues, so the cheap and rich venues stand out per row. **Widest spread** is the highest venue rate minus the lowest: the raw edge before fees. Click a row for its funding history chart (pick the window). Filter by name or by asset class (crypto, or stocks, indices and metals).

Markets → Price matrix shows each asset's mid price per venue. **Dispersion** is highest minus lowest price in bp of the average. A large value usually means the venues list different contracts or a thin book, not free money.

Markets → Hyperliquid spot lists Hyperliquid's spot pairs next to the funding on the perp of the same asset. **Perp funding / yr** is what a long-spot, short-perp pair would collect; **Perp vs spot** is the perp's premium in bp.

Opportunities → Cross-venue funding: the same asset on two venues, long where funding is cheap, short where it is rich. Price moves cancel; the funding difference is the carry. **Opportunities → Spot-perp carry:** long Hyperliquid spot, short the Hyperliquid perp.

How each row is computed, per hour, for a long leg L and a short leg S:

$$edge = f_S - f_L \qquad cost = 2 \times (fee_L + fee_S)$$

$$net(H) = edge \times H - cost \qquad breakeven = cost \div edge$$

Column	Meaning
Long / Short	The leg you buy (cheaper funding) and the leg you sell (richer funding), with each rate
Gross spread / yr	Short funding minus long funding, annualised: what the pair earns before fees
Round-trip fees	Opening and later closing both legs, in bp of one leg's notional
Net / yr	Earnings over the holding period after fees, annualised
Breakeven	Hours of today's spread needed to pay the fees; "no fees" when both legs are maker-free
Price gap	Short price minus long price in bp. Not counted in net: it is price risk, not carry
6 h mean	Average spread over the persistence window from stored history, and the share of minutes it was positive. This is what the strategies test
Thinner leg 24h vol	Volume of the less traded leg
Check	Warnings: thin market, off-hours rules (stocks, metals), extreme rate, large price gap

Controls at the top: **Hold for** (the holding period H that spreads the fees; default 72 h), **Min 24h volume per leg** (default $100,000), and maker or taker fee assumption. Contracts priced more than 5% apart are dropped as different instruments. Click a row for the leg-by-leg detail panel.

How to read it: a high **Net / yr** with a short **Breakeven** and a solid **6 h mean** is worth attention. A high Net / yr with a weak 6 h mean is usually a spike that will be gone before the fees are paid back.

6 Strategies: how the bot decides

Both strategies ship stopped and only open spreads that have lasted: by default the funding spread must have averaged at least 15%/yr (funding arb) over the last 6 hours and been positive 80% of the time. Start one under **Strategies → Funding arb** or **Spot carry** with **Start strategy**; **Stop strategy** stops new entries but leaves open pairs open and watched.

Every 30 seconds a running strategy does this:

1. **Check exits** on its open pairs (below).
2. **List candidates:** spreads above the entry level now, with a taker-fee breakeven within the limit, enough volume on both legs, on venues that are not off or halted, and not already held (by the bot or by you).
3. **Persistence test** from stored history, minute by minute: mean spread over the window at or above the entry level, positive often enough, enough samples. Only rows marked **yes** under **Trades?** pass.
4. **Open the best one** if fewer pairs are open than the cap. Size per leg = 98% of the smaller of the per-pair and per-order caps (default about $245). A preflight checks both legs against the guard first; if either would be refused, the pair is skipped for 30 minutes and nothing is sent.
5. **Execute maker first:** each leg is posted post-only at the best price and re-priced every 8 s, up to 3 times, then finished with a market (IOC) order if still unfilled.
6. **At the cap, rotate:** if rotation is on and a clearly better pair exists, the weakest pair is swapped when the gain over 72 h beats the switching fees.

A pair closes on the first of four exit rules:

- **Spread faded:** the last hour's average spread is below the exit level (hysteresis: exit sits below entry so pairs do not flap).
- **Held too long:** older than the maximum holding time; it is reconsidered fresh afterwards.
- **Legs diverged:** the two legs' prices moved apart by more than the limit since entry, a sign of a broken hedge.
- **Rotated** into a better pair (funding arb only by default).

Every rule is editable under the strategy's **Rules** panel and takes effect at the next check. Defaults:

Rule	Funding arb	Spot carry
Enter at funding spread	15%/yr	12%/yr
Exit below (1 h average)	3%/yr	0%/yr
Persistence window	6 h	6 h
Positive at least	80% of samples	90% of samples
Minimum samples (one per minute)	60	60
Taker-fee breakeven at most	24 h	48 h
Close after	168 h (7 days)	336 h (14 days)
Close if leg prices diverge by	150 bp	100 bp
Minimum 24h volume per leg	$250,000	$250,000

Rule	Funding arb	Spot carry
Rotate into a better pair	on	off
Allow stocks, indices and metals	off	off

The candidate table on each strategy page shows why a spread is not trading yet, for example "only 12 samples (need 60)" or "6h mean 9.4%/yr, positive 71% of the time". Pairs in their 30-minute cooldown and legs already held are left out of the table. When nothing trades for hours, that is the normal state: spreads that survive fees and persistence are rare. Stocks, indices and metals stay off until their off-hours rules and position fees are modelled.

7 Placing orders yourself

Orders & positions → New order sends your own orders to any venue in paper or testnet mode. They go through the same guard as the bot, so the kill switch, venue modes, price sanity and size caps all apply, and every order is logged.

A single order, step by step:

1. **Venue:** only venues set to paper or testnet are listed, with their mode in brackets. Change modes under Risk → Limits.
2. **Market type** (Hyperliquid only): Perp or Spot.
3. **Market:** the dropdown lists every tradable market on that venue, busiest first, with price and 24-hour volume; the number in brackets is how many there are. Type in **Search** to narrow it (for example "SOL").
4. **Side:** Buy / long or Sell / short.
5. **Order type:**
   - **Market** fills now at the best price, up to 20 bp beyond it. If the price runs further before the order lands, it is not filled at all (you see "not filled"), never filled at a bad price.
   - **Limit** fills at your price or better. If your price already crosses the book it fills now as taker; otherwise it rests on the book as maker until filled or cancelled.
   - **Post-only** only ever rests as maker (lower fee). The ticket blocks one that would cross.
6. **Amount** in USD of notional or in units of the coin (switch with the USD / coin buttons). USD is converted at the current mid.
7. **Limit price** (limit and post-only): type it, or press **Bid**, **Mid** or **Ask** to fill it from the live book.
8. **Reduce-only:** tick it to only shrink or close a position, never open or flip one. Reduce-only orders pass even while the kill switch is on.
9. **Review order** shows exactly what will be sent: side, size, venue and mode, price, notional, maker or taker, estimated fee. Nothing has been sent yet; **Back** returns to editing.
10. **Place buy order / Place sell order** sends it. The result appears below: filled (size, price, fee), resting (see Open orders), or refused with the reason.

The panel on the right updates every 2 seconds and streams the market live while the page is open: bid, ask, spread, book age, funding per year, maker and taker fee, your position, account equity, margin in use, and for your order the notional, the margin it needs (notional ÷ leverage; full notional for spot) and the estimated fee.

A hedged pair (lower half of the page): pick the long venue (and Perp or Spot on Hyperliquid), the short venue, an asset listed on both (the dropdown only offers those), and USD per leg. **Review pair**, then **Open pair**. Both legs are worked maker first, then crossed, exactly like the bot's pairs. The pair is recorded with strategy "manual", so the imbalance and liquidation brakes watch it and the strategies will not open another pair on the same legs. There is no persistence test: you are deciding.

Cancel and close:

- **Open orders → Cancel** removes a resting order. Anything already filled stays filled.
- **Positions → Legs → Close** closes one market's whole position with a reduce-only market order (taker fee).
- **Positions → Pairs → Close** closes both legs of a pair, maker first.

What gets refused, and why:

Message contains	Cause	What to do
"is off" or "is set to 'off'"	The venue is in off mode	Set it to paper or testnet under Risk → Limits
"kill switch"	Kill switch engaged	Tick Reduce-only, or release the kill switch
"through mid"	Limit price more than 70 bp past the market (20 bp slippage + 50)	Use a price nearer the book, or a market order
"would cross the book"	Post-only price at or through the other side	Use Limit, or move the price
"cap"	Order, venue or whole-book size cap	Make it smaller, or raise the cap under Risk → Limits
"cannot sell more spot than held"	Spot cannot be shorted	Sell at most what you hold; spot sells are always reduce-only
"insufficient paper collateral"	Not enough free collateral on that venue	Smaller order, or reset the paper account
"leg of open pair"	Closing one leg would leave a pair one-legged	Close the pair, or trim with a reduce-only order
"halted" or "Book is 40s old"	Stale feed or repeated order errors on that venue	See Risk → Brakes; resume the venue once fixed

One thing to know: positions are netted per market. If you buy BTC on Hyperliquid by hand while the bot also holds a BTC leg there, the venue shows one combined position. Keep manual trades on markets the bot is not using, or use reduce-only orders to trim exactly what you added.

8 Orders and positions

Positions shows what is held now; Open orders what is resting; Trade history what happened.

Positions has four headline tiles: **Equity, all venues** (paper balances plus testnet accounts at mark), **Paper P&L since start** (after fees and funding), **Funding on open legs** (received minus paid) and **Open pairs**. Below them:

- **Pairs:** each hedged pair with its strategy (funding arb, spot carry or manual), long and short venue, size per leg, entry spread, how long it has been held, and **Close** (both legs, maker first, then market). Status runs opening → open → closing → closed, or failed if neither leg filled.
- **Legs:** every open position on every venue: signed size (+ long, – short), notional, entry and mark price, **Price P&L** (offset by the other leg in a pair), **Funding** so far, **Fees**, **To liquidation** (red under 30%) and **Close** for that leg.
- **Accounts:** per venue, the mode, equity, cash, margin used and P&L. **Reset paper** returns a paper account to its starting balance; it is allowed only with no open positions.

In a funding pair, judge the result by **Funding minus Fees** across both legs. Price P&L on one leg is expected and should roughly cancel against the other leg.

Open orders lists orders resting on a venue now: time, venue and mode, asset, side, type (post-only or IOC, plus "reduce"), size, limit, filled, average price, fee and status, with **Cancel** per row. The bot's own orders live for seconds before they fill, are re-priced or cross. Your limit orders stay until they fill or you cancel them.

Trade history has three views: **Pairs** (closed pairs with the reason in **Why closed**), **Orders** (every order that reached a venue, including ones the venue rejected, with the reason; orders the guard refused before sending show where you placed them: on the ticket, or in the strategy's status note) and **Funding** (each payment received or paid per leg). Maker fills show their lower fee here, so you can see how often maker-first works.

9 Risk: limits, brakes and the kill switch

Risk decides where IT45 may trade and how much, and stops it automatically when something goes wrong. Every order, the bot's or yours, passes a fail-closed guard in this order: kill switch → venue mode → venue halted → book freshness and price sanity → size caps. Any failed check refuses the order.

Risk → Limits

Venue modes, one per venue:

Mode	What happens	Available on
Off	Not traded. Market data still updates	All
Paper	Simulated fills on the venue's live prices with play money (default)	All
Testnet	Real signed orders on the venue's test exchange with play money; needs a key in `.env`	Hyperliquid, Bulk, Meridian

The **Connector** column says whether a real connector exists and whether its key was found. If a testnet connector fails to start, the reason shows there and every order to that venue is refused; it never falls back to paper silently.

Setting	Default	Meaning
Per order	$250	Largest single order
Per pair leg	$250	Notional of each leg when a pair opens (pairs use 98% of the smaller of this and per order)
Per venue, gross	$600	Longs plus shorts on one venue
Whole book, gross	$2,000	All venues together
Open pairs at once	3	The strategies stop opening at this many, counting your manual pairs; your own pairs are not blocked by it
Leverage per leg	2x (1 to 10x)	Funding arb earns nothing extra from leverage; higher only moves liquidation closer
Paper starting balance per venue	$1,000	Applies when a paper account is reset
Post-only attempts before crossing	3	Maker re-prices before a market order finishes the leg
Re-price every	8 s	Wait before moving a post-only order to the new best price
Max distance for market orders	20 bp	How far past the best price a market order may fill
Cross the spread when post-only does not fill	on	Off: the leg stays partly filled and the imbalance brake handles it

Press **Save limits** after editing. Invalid values are refused with the reason (for example an exit level above the entry level, or leverage above 10x).

Risk → Brakes. Four brakes run every cycle, independent of the strategies:

Brake	Trips when (default)	Then
Stale feed / API errors	A venue's prices are older than 30 s, or 3 order errors in a row	The venue is halted: no new orders there, closing orders still pass. A stale halt lifts itself when prices are fresh; an error halt needs **the Resume button for that venue** on this page
Daily loss	Total equity falls 3% of starting capital below its level at 00:00 UTC	Everything is closed, the kill switch engages, strategies stop. It re-arms at 00:00 UTC
Hedge imbalance	A pair's legs differ by more than 10% for longer than 45 s	The heavier leg is trimmed; a lone leg is closed
Liquidation distance	A leg is within 20% of its liquidation price	The pair is halved

The page also shows **P&L today (UTC)** against the limit, the **Halted venues** with their reason, and the state of the daily loss brake (armed or tripped).

Risk → Kill switch is the emergency stop.

1. **Close everything** asks for confirmation first.
2. **Yes, close everything now** cancels every order and closes every position on every venue with market orders, stops both strategies and engages the kill switch. While it is on, only reduce-only (closing) orders pass, including yours.
3. **Release kill switch** allows new orders again. Strategies stay stopped until you start them.

Use it when something looks wrong and you do not know why. Closing costs taker fees plus the spread.

If the internet drops. After 30 seconds without fresh prices every venue is halted and the guard refuses all orders, closing ones included, because it cannot price them; a red banner says so. Open positions stay on the venues, hedged, and IT45 cannot act on them until the connection returns. An order sent just as the line dropped is marked *unknown*: IT45 sends nothing more for that leg, and once the venue answers it asks what became of the order, cancels it if it is still resting and books any fill. A disconnection never engages the kill switch and never closes positions: while a venue is unreachable, and for 60 seconds after it returns, no brake or strategy exit acts on a pair with a leg there, and the daily-loss check pauses whenever an account balance cannot be read, so missing data is never counted as a loss or a missing leg. After that the brakes check the real positions, repair any one-sided pair, and the strategies carry on. Every request to a venue gives up after 10 seconds instead of hanging. A laptop that sleeps is the same as an outage; keep it awake while IT45 runs (`powercfg /change standby-timeout-ac 0`).

10 Settings, logs and keys

Settings changes how IT45 reads the markets; keys are never typed into the dashboard but put in `.env`.

Settings → General

Setting	Default	Meaning
Poll every	5 s	How often every venue's REST API is read (minimum 2: Bulk allows 5 requests per second)
Store a snapshot every	60 s	How often prices and funding go into history. The persistence test counts minutes, so keep it at 60 or below
Default holding period	72 h	What the Opportunities tables start with
Default fee assumption	maker	Maker or taker fees in the Opportunities tables
Default minimum 24h volume per leg	$100,000	Volume floor the Opportunities tables start with

Settings → Fees: the base maker and taker fee per venue, with two boxes for yours. Fill them only if your volume tier is better; leave empty to use the base. Opportunities and paper fills use them immediately.

Settings → Venues: switch a venue's market feed on or off. A venue with its feed off cannot be traded (its prices go stale and the stale brake halts it).

Settings → API keys: per venue, the key type, whether that key type can withdraw (IT45 only uses keys that cannot), and whether this build can trade the venue and found your key.

Logs → Events: feed failures, recoveries and settings changes, newest first. Filter by level (info, warning, error).

Setting up testnet keys. Copy `.env.example` to `.env` in the IT45 folder, fill in the lines below, and restart IT45. Then set the venue to testnet under Risk → Limits. Never put a main wallet's private key in `.env`; full steps are in `docs/03_keys.md`.

Venue	How to get the key	`.env` lines
Hyperliquid	On app.hyperliquid-testnet.xyz: claim test USDC from the faucet, then More → API, create and approve an API wallet, copy its private key once	`IT45_HL_AGENT_KEY` (API wallet key), `IT45_HL_ACCOUNT_ADDRESS` (your main wallet address, not a key)
Bulk	Run `.\.venv\Scripts\python.exe scripts\bulk_testnet_wallet.py`: creates a testnet-only wallet, writes the key and claims 1,000 test USD (again every 24 h to top up)	written for you: `IT45_BULK_SIGNER_KEY`, `IT45_BULK_ACCOUNT`
Meridian	Fund a testnet-only wallet at faucet.meridiantest.net; put its key in `.env` as `IT45_MERIDIAN_OWNER_KEY`; run `scripts\meridian_link_signer.py`; then delete the owner key line	written for you: `IT45_MERIDIAN_SIGNER_KEY`, `IT45_MERIDIAN_ACCOUNT`
Arcus, Reya	No public testnet yet	paper only

Check a connector live with `.\.venv\Scripts\python.exe scripts\testnet_smoke.py bulk` (or `meridian`, `hyperliquid`): it rests an order 10% under the market, cancels it, opens about $20 with a market order and closes it, printing every step.

11 Paper versus testnet: what the results prove

Paper shows whether the strategy would earn on real prices; testnet shows that the order plumbing works. Neither is real money, and neither alone proves the other.

	Paper	Testnet
Prices	Live mainnet prices and funding	The test exchange's own play market
Orders	Simulated inside IT45	Real signed orders to the venue's test exchange
Proves	Whether spreads earn after fees and funding	Signing, order types, fills, cancels, positions, brakes, kill switch on the real API
Does not prove	That the venue accepts and fills the orders	Profit: test-market prices and funding are not real
Money	$1,000 per venue, reset any time	Faucet play money

In testnet mode the strategies still choose pairs from mainnet funding, because test-market rates mean nothing.

How paper fills work:

- A **market (IOC)** buy fills its full size at the ask if the ask is within the limit, taker fee; otherwise nothing. There is no depth model, which flatters large orders; the caps keep orders far below each venue's volume.
- A **post-only** buy at price P is rejected if P is at or above the ask. Otherwise it rests. With the live stream on, it fills by a queue model: it waits behind the size already shown at that price, taker sells at that price use up the queue first, and a sell printed below P fills it at once. Without the stream, it fills only once the ask trades below P. Sells mirror this.
- **Funding** accrues continuously on every perp leg (longs pay positive rates) and is booked once a minute. Spot legs pay none.
- **Margin:** perp legs need notional ÷ leverage, spot legs the full notional. **Liquidation** is estimated per leg, isolated style, which is conservative.

Not modelled: order-book depth, cancellations ahead of you in the queue, partial fills, venue outages mid-order, hourly settlement timing, Meridian position fees and Arcus off-hours bands. Treat paper P&L as a good estimate for small sizes, not as a promise.

12 Daily routine and first week

A daily check takes two minutes; the first week builds the history the strategies need and proves each piece once.

Every day:

- [ ] Top bar: five green venue dots, the mode pill says what you expect, no red kill-switch pill.
- [ ] Risk → Brakes: no halted venues; P&L today well inside the limit.
- [ ] Orders & positions → Positions: every pair has both legs, no leg red under To liquidation, funding on open legs above fees.
- [ ] Trade history → Pairs: read **Why closed** for anything closed since yesterday.
- [ ] Logs → Events: no repeating errors.

First week:

- [ ] Day 1: start IT45 with `.\run.ps1` and leave it running. Snapshots start at once; the strategies need at least 6 hours of history.
- [ ] Day 1: walk every page with hover help on.
- [ ] Day 1: place a small manual order on paper (New order, $50 market buy), see it under Positions, close it.
- [ ] Day 1: place a limit order below the bid, find it under Open orders, cancel it.
- [ ] Day 1, evening: open Strategies → Funding arb and check the candidates table; rows marked yes would trade.
- [ ] Day 2: start Funding arb on paper and watch the first pairs open and close.
- [ ] Day 2: test the kill switch once on paper; release it and restart the strategy.
- [ ] Days 2–3: set up the Bulk testnet key (one script), switch Bulk to testnet and repeat the manual order test.
- [ ] Days 3–7: compare Funding minus Fees across closed pairs; decide whether the entry and exit levels need tuning.
- [ ] End of week: tell Claude what you saw (pairs opened, closed, P&L, anything odd) before deciding on M3.

13 Troubleshooting

Most problems show their cause on the page where they happen; this table covers the ones that do not.

Symptom	Likely cause	Fix
`.\run.ps1` is refused	PowerShell blocks scripts	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, once
Red "Install failed" and IT45 does not start	A package did not install	Send the pip error text to Claude
Browser says the page cannot be reached	IT45 not running, or another program on port 8045	Check the PowerShell window; or start with `.\run.ps1 --port 9000`
"address already in use"	IT45 is already running in another window	Use the open window, or close it with Ctrl+C first
A venue dot is amber or red	That venue's API is slow or down, or rate-limiting	Wait; it recovers by itself. Overview → Venue health shows the error
Red banner "No venue is answering"	Internet connection down, or the computer slept	Nothing to do in IT45: new orders are paused and resume by themselves. Restore the connection
Amber banner "orders sent without a reply"	The connection dropped while an order was in flight	Wait: once the venue answers, IT45 asks what became of each order, cancels any left resting and logs the result under Logs → Events
Red banner "IT45 is not responding"	IT45 stopped (window closed, crash) while the page was open	Start it again with `.\run.ps1`; open pairs are picked up from the database and the venues
Strategy running but no pairs open	Normal most of the time: nothing passes entry and persistence	Read the candidates table; a new install needs 6 h of history first
Pair shows only one leg	One leg filled, the other did not	The imbalance brake repairs it within about 45 s; check Positions again
Venue listed under Halted venues	Stale feed (lifts itself) or 3 order errors in a row	Read the reason; fix it, then press its Resume button on Risk → Brakes
Testnet venue refuses every order	Key missing or connector failed to start	Risk → Limits → Connector column shows the reason; check `.env`, restart
Everything closed and only closing orders pass	Daily loss brake or kill switch	Risk → Brakes shows which; release the kill switch when you are ready
Reset paper is refused	The account still holds positions	Close them first under Positions
New order says "not filled"	The price moved more than 20 bp before the market order landed	Try again, or use a limit order
A page looks out of date after `git pull`	Browser cache	Press Ctrl+F5

For anything else, note the time, the page and the exact message (hover it if needed) and send them to Claude. `data/it45.db` holds the full audit trail of every order decision.

14 Glossary

Term	Meaning
Basis point (bp)	0.01%. 100 bp = 1%
Bid / ask / mid	Best price buyers pay / sellers accept / halfway between
Breakeven	Hours of today's spread needed to pay the round-trip fees
Carry	Income from holding a position, here funding received
Delta-neutral	Long and short the same asset in equal size, so price moves cancel
Funding	Hourly payment between longs and shorts that keeps a perpetual near spot. Positive: longs pay shorts
Funding spread (edge)	Short-leg funding minus long-leg funding: what a hedged pair earns
Gross / net	Before / after fees
Guard	The fail-closed check every order passes before it is sent
Hedged pair	One long leg and one short leg of the same asset and size
Hysteresis	Exit level below entry level, so a pair does not open and close on every wiggle
IOC	Immediate-or-cancel: fills now up to a price, the rest is cancelled. Used for market orders
Leg	One side of a pair: one position on one venue
Leverage	Notional divided by margin. Higher leverage needs less collateral and moves liquidation closer
Liquidation	Forced close by the venue when margin runs out
Maker / taker	Maker: your order rested on the book (lower fee, often 0). Taker: it crossed the spread
Margin	Collateral reserved for a position
Mark price	The venue's price for valuing and liquidating positions
Notional	Size times price, in USD
Oracle	External index price a perpetual is anchored to
Paper	Simulated trading on live prices with play money
Perpetual (perp)	Futures contract with no expiry, kept near spot by funding
Persistence	How long and how consistently a spread has existed in stored history
Post-only	Limit order that only rests as maker and is refused if it would cross
Reduce-only	Order that can only shrink or close a position
Rotation	Swapping the weakest pair for a clearly better one when at the pair cap
Spot	The asset itself, no funding, no leverage
Spread (book)	Ask minus bid
Testnet	The venue's test exchange with play money and real signed orders

15 What this build cannot do yet

This is build M2.x: no real money can be traded, by design.

- **No mainnet.** M3 adds a per-venue Arm switch. Before any venue is armed you confirm in writing that its terms allow your country of residence, and mainnet keys are trade-only agent or linked-signer keys without withdrawal rights.
- **No alerts yet.** Telegram, email and Windows notifications come with M3; until then, look at the dashboard.
- **Arcus and Reya are paper only** until either publishes a testnet host.
- **Meridian testnet** signing is verified; a full order cycle on a funded account has not been run yet (run `testnet_smoke.py meridian` after setup).
- **Laptop only.** IT45 runs while your PowerShell window is open; VPS packaging (Docker) is part of M3.
- **Stocks, indices and metals** are shown but not traded by the strategies: their off-hours rules and position fees are not modelled.
- **Guard refusals of bot orders** are stored in the database audit log but have no page of their own yet.

The hover help always matches the running build; this guide describes the build of 4 October 2026 (commit 9e89ea9).